

# CIRCUITS: HOW DOES A TORCH WORK?



Let's Discover Light Together!



LET'S WONDER

# What Do We Use in a Power Cut?

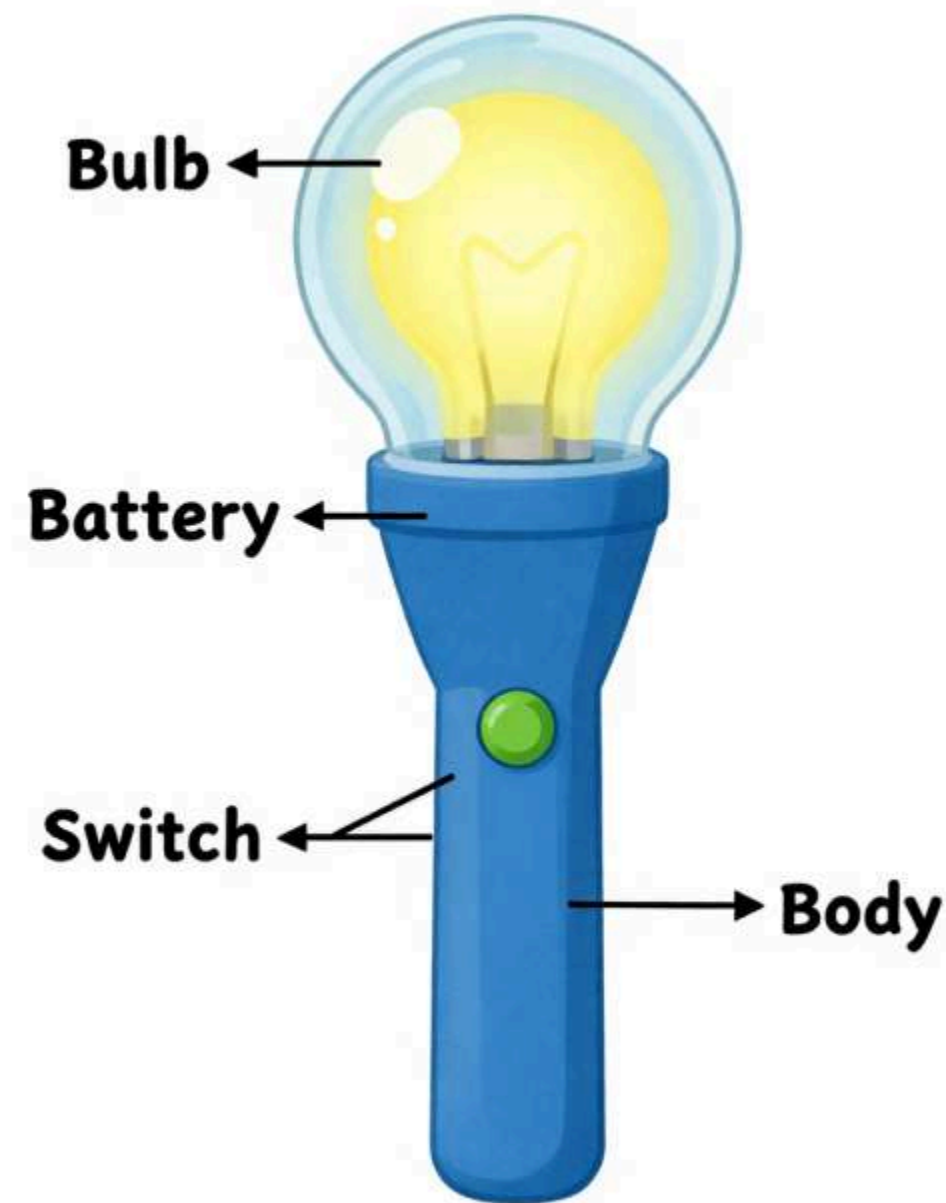
Imagine it's nighttime. Suddenly, all the lights go off! The room becomes very dark. You can't see your toys or your books. What would you look for? What helps us see in the dark? Think about it!

Think!

When the power goes off,  
what do we reach for?



# Look inside a torch. Can you name all the parts?



## Battery

This stores all the energy inside the torch.

## Bulb

The part that glows bright when electricity flows through it!

## Switch

Push it to turn the torch on and off.

## Body

The outer case that holds everything together safely!

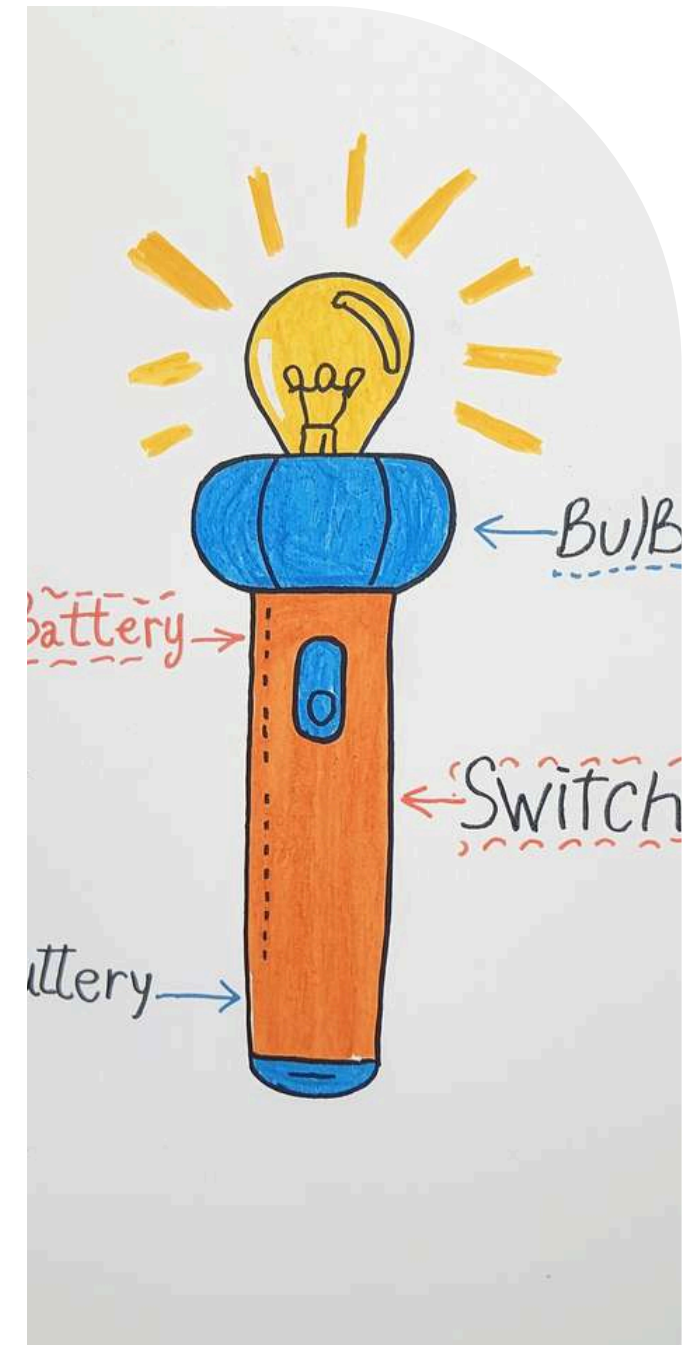
# Let's Draw Our Torch!

Follow these easy steps to draw your own torch!

- Draw a long rectangle for the body
- Add a small circle at the top for the bulb
- Draw a tiny rectangle on the side for the switch
- Label these 3 parts: Battery, Bulb, Switch

Colour your torch:

- Saffron-Orange for the body
- Peacock Blue for the switch
- Sunshine Yellow for the light





LET'S LEARN!

# How Does Electricity Move?

Think of electricity like children running on a track. They need a complete path to keep moving! If the circle breaks, the children stop. Electricity works the same way – it needs a complete loop to flow.

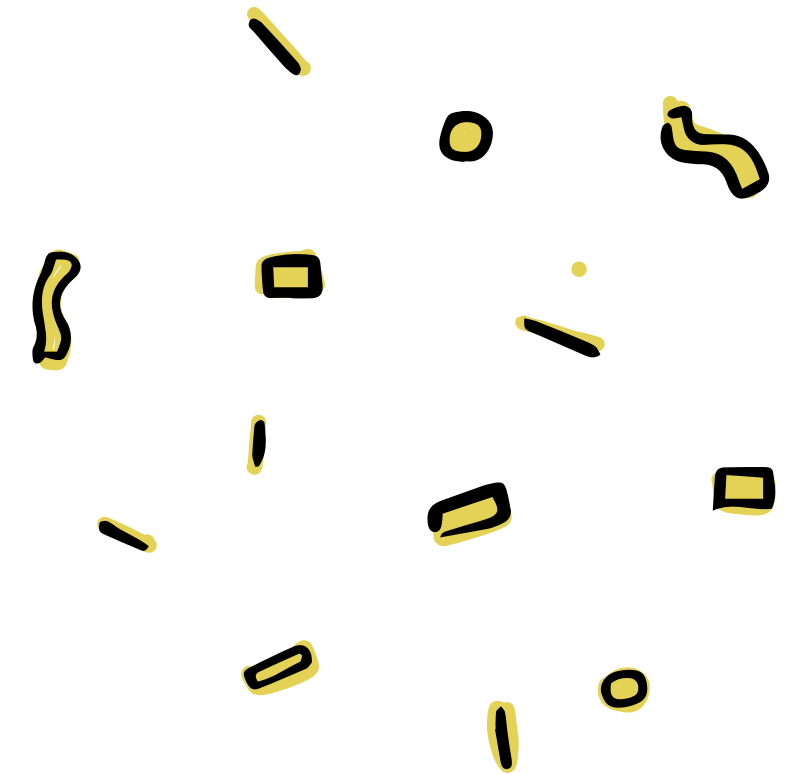
## Key Idea

Electricity moves in a circle. Like running round a track!





## What Does the Switch Do?



**\* Switch ON**  
The path is complete! Electricity flows through and the bulb glows bright like a little sun.

**\* Switch OFF**  
The path is broken! Electricity cannot flow, so the bulb stays dark and sleepy.

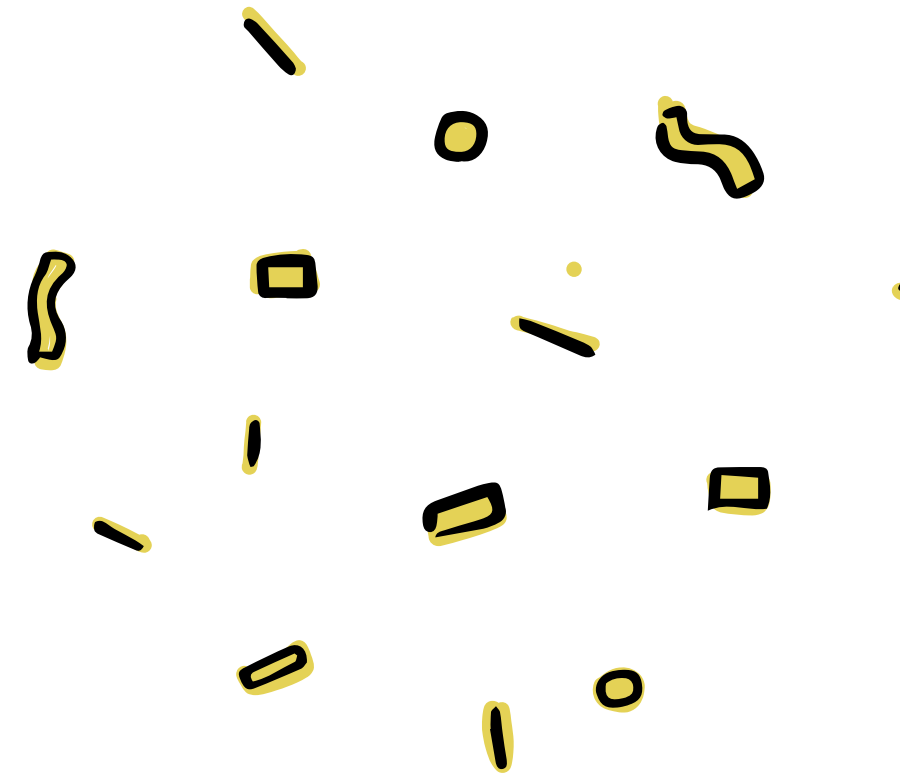
# Battery – Our Energy Source

## Energy Storage

\* The battery stores energy inside it, just like your tiffin box stores yummy food for later! When you need light, the battery shares its energy.

## Energy Flow

\* The battery pushes electricity around the torch circuit. The energy travels from the battery, through the wires, to the bulb, and back again!

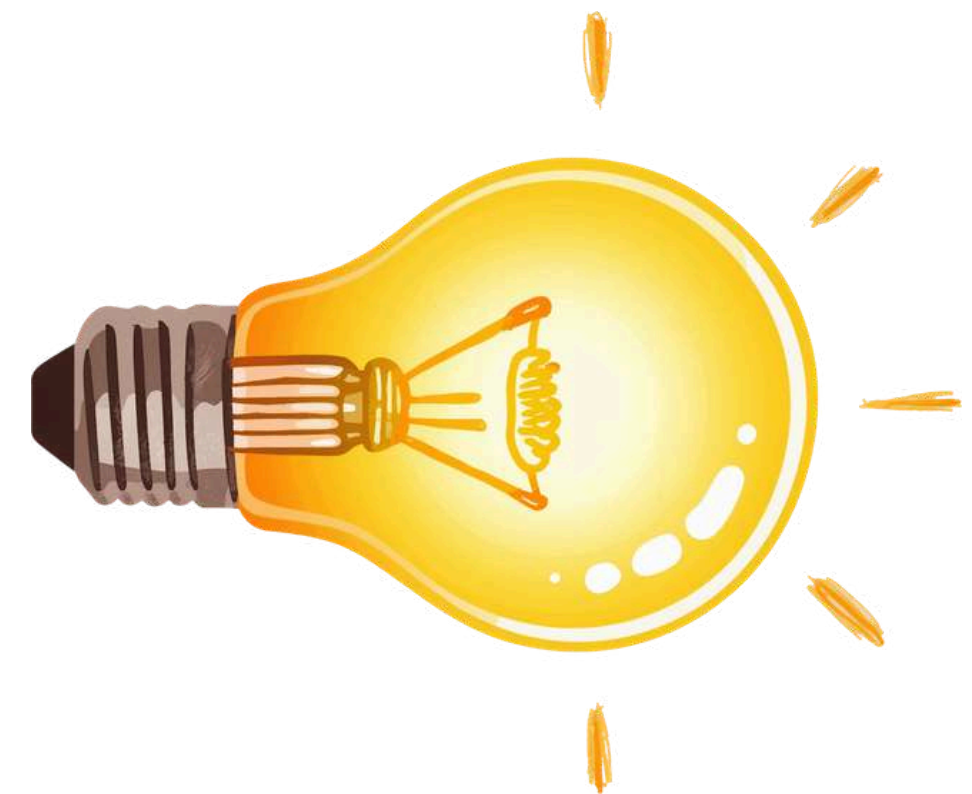


SEE IT GLOW!

# The Bulb Glows!

The bulb is the part of the torch that makes light. Inside the bulb is a very thin wire called a filament. When electricity flows through this tiny wire, it gets so hot that it glows bright! That's how we get light from our torch.

When electricity reaches the bulb, the thin wire inside gets very hot and starts to glow!





# What Happens if...

Let's see what happens when we change things in our torch circuit!

Sometimes the torch works, and sometimes it doesn't. Can you guess why?

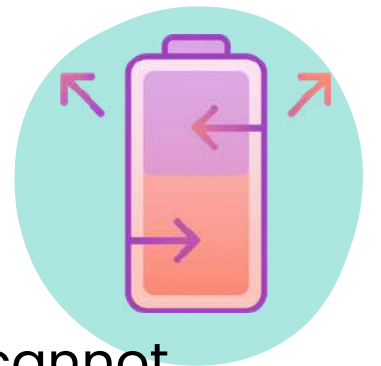
## No Battery

Without a battery, there is no energy. The torch cannot glow!



## Battery Backwards

If the battery is put in the wrong way, electricity cannot flow. No light!



## Circuit Complete

When everything is connected right and switch is ON – the torch glows bright!





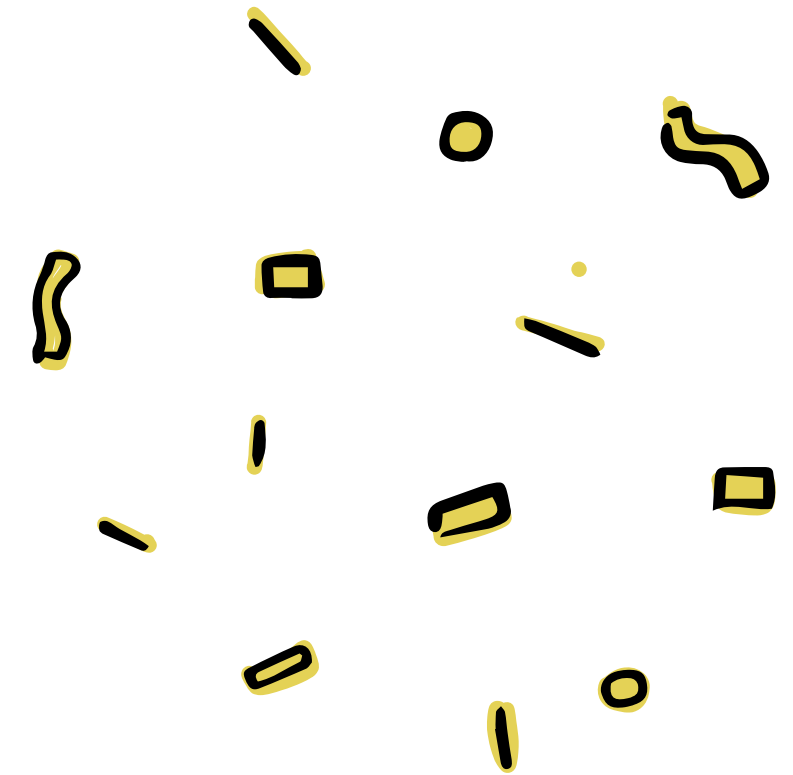
## Torches help us every day in India!

### \* Power Cuts at Home

Every Indian family keeps a torch ready! When the lights go off, we grab our torch to find our way around the house safely.

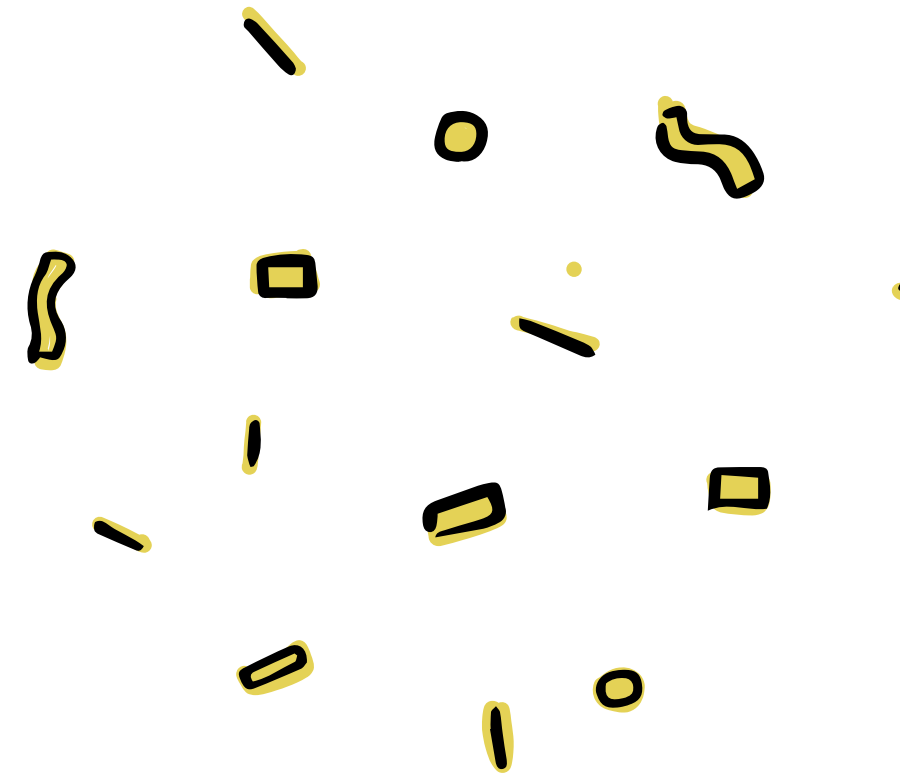
### \* Night Workers

Watchmen use torches to keep our homes safe at night. Street vendors use torches to show their goods after dark.





**Torch light can change color with colored paper!**



**\* Try This at Home!**

Hold colored paper in front of your torch. Red paper makes red light! Blue paper makes blue light! What colors can you make?

**\* Color Magic!**

Remember our color lessons? White torch light has all colors hiding inside! Colored paper lets only one color through. It's like magic!

# Quick Recap

Let's remember what we learned about torches!  
Electricity moves in a circle like running around a track. If the circle breaks, the light stops.

## Name the Parts

Battery, Bulb, Switch, and Body work together to make light!



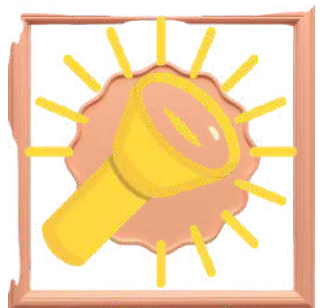
## Switch Opens & Closes

The switch opens and closes the path for electricity. ON means light, OFF means no light!



## Torches Help Us!

Torches help us see in the dark during power cuts.





# Thank You & Keep Exploring!



Keep shining bright like a torch! Do you have any questions? Try drawing your own torch and share it with your friends!



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